

Resource use efficiency and farm productivity gaps of smallholder dairy farming in North-west Michoacán, Mexico



José Cortez-Arriola^{a,1}, Jeroen C.J. Groot^{a,*}, Ricardo D. Améndola Massiotti^{b,2}, Johannes M.S. Scholberg^{a,1}, D. Valentina Mariscal Aguayo^{b,2}, Pablo Tittonell^{a,1}, Walter A.H. Rossing^{a,1}

^a Farming Systems Ecology Group, Wageningen University, P.O. Box 563, 6700 AN Wageningen, The Netherlands

^b Graduate Program in Animal Science, Chapingo University, Km. 38.5 carretera Mexico-Texcoco, CP 56230, Texcoco, Estado de Mexico, Mexico

ARTICLE INFO

Article history:

Available online 7 December 2013

Keywords:

Farm yield gap
Farming intensity
Herd structure
Milk production
External dependence
Nutrient balances

ABSTRACT

Smallholder dairy farms that intensify production risk resource degradation and increased dependence on external feeds and fertilizers due to lack of knowledge and appropriate technology, which undermines farm productivity and profitability. Here we analyze underlying causes at farm level of such process through an integrated analysis at the farm scale by assessing current resource use efficiency for grazing-based dairy farming systems representative of NW Michoacán, Mexico. Whole-farm yield gaps were quantified by comparing current farms to virtual reference farms that have the same farm surface area but improved farm management. Productivity of reference farms was calculated by assuming best crop production practices (as observed within the set of case study farms) and improved herd management. Three family-based (FB) and three semi-specialized (SS) dairy systems spanning three levels of intensification in terms of density of livestock units (LU): extensive (E, <0.8 LU ha⁻¹), medium-intensive (M, between 0.8 and 1.2 LU ha⁻¹), and intensive (I, >1.2 LU ha⁻¹) were monitored during one year (rainy and dry seasons) to assess productivity and resource use efficiencies. Milk production was generally low and variable (2.2–4.3 Mg milk cow⁻¹ lactation⁻¹, and 0.6–5.8 Mg ha⁻¹) due to high incidence of mastitis, a large fraction of non-productive animals in the herd and inefficient reproduction management. During the dry season, grazing areas provided insufficient metabolizable energy, and milk production was sustained through increased use of concentrates (from 310 g kg⁻¹ DMI in rainy season to 454 g kg⁻¹ DMI⁻¹ in dry season of the herd) and conserved forage. All farms had positive nitrogen, phosphorus and potassium balances, averaging 75 ± 16, 15 ± 6, and 19 ± 6 kg ha⁻¹, respectively. Nutrients in animal excreta were mostly not recycled on the farms but lost to the environment, and nutrient surpluses increased with livestock density. The reference farms exhibited an attainable milk yield of 2.7 Mg ha⁻¹ on the basis of full feed self-supply, and 4.2 Mg ha⁻¹ when the crude protein limitation in the ration was lifted. Compared to the reference farm actual milk yields were on average 78.4% lower on FB farms and 57.9% lower on SS farms. The underlying causes of the farm yield gap differed between farms and were due to sub-optimal areas of forage maize, low forage and forage maize productivity and deficient herd management. We conclude that the farm yield gap analysis was effective in identifying the major shortcomings in management of the dairy farming systems and enabled formulation of change avenues for farm reconfiguration focusing on combined improvements in crop, feed and herd management and recycling of nutrients through manure management.

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1. Introduction

Although smallholder farms produce a large share of the world's food supply and their production systems could potentially be diverse and sustainable, market and policy developments force them to intensify their production in order to compete with larger specialized farms (Kiers et al., 2008; Herrero et al., 2010). This often leads to an 'intensification-trap': in order to increase productivity small farmers intensify their production systems by increasing livestock density and inputs, but inadequate

* Corresponding author. Tel.: +31 317 485924/317 481191; fax: +31 317 481203.

E-mail addresses: jose.cortezarriola@wur.nl (J. Cortez-Arriola), jeroen.groot@wur.nl (J.C.J. Groot), r_amendola@yahoo.com (R.D. Améndola Massiotti), johannes.scholberg@wur.nl (J.M.S. Scholberg), valmara@hotmail.com (D. Valentina Mariscal Aguayo), pablo.tittonell@wur.nl (P. Tittonell), walter.rossing@wur.nl (W.A.H. Rossing).

¹ Tel.: +31 317 481191; fax: +31 317 481203.

² Tel./fax: +52 (595) 9521621.

management results in larger nutrient surpluses or losses, farm resource degradation and strong dependence on external feeds and fertilizers.

The intensification trajectories of dairy farms in the northwest of Michoacán (Mexico) are a point in case. In this region where dairy production is an important economic activity, nutrient surpluses from livestock systems resulting in nutrient runoff and leaching are considered to be a major cause of negative environmental impacts on the 'Lerma – Chapala' basin. In Lake Chapala, the largest fresh water lake of Mexico, high concentrations of N and P compounds (2.23 and 0.57 mg l⁻¹, respectively) hamper the use of water resources for human consumption and irrigation (Silva et al., 2002; Ramírez et al., 2007). In the lowest part of the basin, continuous use of saline groundwater has increased the salinity and sodicity of the soil, limiting crop production and impacting soil management (Silva-García et al., 2006).

In addition to these environmental impacts, profitability of local production systems is cause for concern, mainly related to the high feed costs and the low milk productivity in the systems (Espinoza-Ortega et al., 2005; Cortez-Arriola et al., submitted for publication). Most livestock feeding systems in Mexico depend on the use of concentrates, conserved forage, and grazing land. In Mexico forage production during late spring and summer (the rainy season in most of the country) contributes between 60% and 79% to the total annual forage production (Améndola et al., 2005; Martínez et al., 2008; Sosa et al., 2008). Farmers address potential periodic forage shortages via the use of largely imported conserved forage (silage, hay and stover) or concentrates (Améndola et al., 2005; Cortez-Arriola et al., submitted for publication). However, these options increase the dependence on external feed resources: 70% of the ration and represents up to 90% of the feeding costs. As a consequence farm profitability is reduced (Espinoza-Ortega et al., 2005; Cortez-Arriola et al., submitted for publication). On the other hand, animal productivity is a key variable to enhance the milk production in Mexico and the profitability of farming. In the period from 2004 to 2010 the average production was only 1.65 Mg milk cow⁻¹ lactation⁻¹ (SAGARPA, 2011). In contrast, in The Netherlands and Italy where feeding is mainly based on own forage production with low contribution from external inputs in feeding costs (ca. 18% of the total costs) production levels were 7.6 and 6.7 Mg milk cow⁻¹ lactation⁻¹ (European Commission, 2011).

Efforts to develop more sustainable intensification trajectories may benefit from interventions based on yield gap analysis at the farming system level. To diagnose differences in performance among farms that operate under similar climatic conditions and natural resource availability, farm yield gap analysis is relevant to identify the most limiting and reducing factors that determine the yields and to find adjusted practices that could contribute to closing the gaps. Usually, yield gap analyses are conducted to determine differences between attainable and actual crop yields in cropping systems (Bhatia et al., 2006; Nin-Pratt et al., 2010; Van Ittersum et al., 2013). To narrow the gap, field crop production practices such as use of fertilizers, establishment of irrigation systems, and genetic improvements may then be proposed. In more complex agro-ecosystems such as mixed crop-livestock systems, a large farm yield gap can originate from shortcomings related to farm design, resource allocation and tactical planning. The resulting imbalances between feed supply and herd size combined with lack of effective herd management practices thus may hamper optimal use of forage resources and prevent farmers from attaining the production potential of livestock.

Here, we present a method to diagnose farming systems by constructing a virtual farm based on best existing local practices and use this as a reference to assess the farm yield gap and the contribution of constraints in the various sub-systems. We quantitatively analyze six dairy farms that represent the variation of dairy farming

in the region of Marcos Castellanos, Michoacán, Mexico (Cortez-Arriola et al., submitted for publication). Furthermore, we assess how farm management affects annual metabolic energy (ME), crude protein (CP), and N, P, and K balances. The objectives of this paper are: (a) to quantify the gap between attainable and current farm milk yields taking a whole farm perspective, (b) to diagnose the major factors that limit productivity of dairy farming systems in NW Michoacán; and (c) to analyze how management affects farm gate nutrient balances and use efficiency of nutrients.

2. Material and methods

2.1. Case study region

The municipality of Marcos Castellanos is located in the Northwest of Michoacán, Mexico, (19° northern latitude and 103° west longitude) at altitudes between 1500 and 2400 m above sea level. According to the classification of Mexican hydrological regions it is located in the upper part of the 'Lerma – Chapala' basin, in the sub-basin 'Chapala', and in the micro-basin 'San José de Gracia'. The area of the municipality is 23,285 ha, representing 0.39% of the State; 86% of the area is grazing land (mainly native rangeland) and 12% is cropland, mostly used for forage maize production (Secretaría de Gobierno, 2010). The climate is classified as temperate with one main rainy season between June and October. Overall precipitation and temperature averages are 798 mm and 18.9 °C, respectively. Chromic and pellic Vertisols are the dominant soil type, in associations with Luvisols, Inceptisols, Phaeozems, and Andosols (SEMARNAT, 2003). The physiography is constituted by slightly sloping hills ending in ravines, and small almost flat areas. Thorny forest is the dominant type of vegetation, followed in importance by broadleaf forest (SAGDER, 2000).

2.2. Selection of farms representative of regional farming systems

During 2007 a survey was carried out in the region, which was used to develop a farm typology presented elsewhere (Cortez-Arriola et al., submitted for publication). From the total of 630 farms listed in the census of the regional association of dairy farmers, 14.6% were randomly selected and interviewed. Multivariate analysis of the survey data resulted in six main farm types, distinguished by the use of family versus hired labor and the intensity of production, in particular cattle densities. Based on these farm types and the national Mexican dairy farm classification (FIRA, 1997; SAGARPA, 2000; Amendola, 2002), six farms were selected for more detailed study. These farms are being referred to as family-based (FBx) or semi-specialized (SSx), where x describes the intensification level based on livestock density as E (extensive, <0.8 LU ha⁻¹), M (medium-intensive, between 0.8 and 1.2 LU ha⁻¹) or I (intensive, >1.2 LU ha⁻¹) (Table 1).

2.3. Farm characterization

In order to characterize each individual farm, close monitoring of system management and agro-ecological parameters took place between July 2009 and June 2010. At the crop subsystem level, information was collected on forage production, quality, utilization, inputs, and production costs. At the animal subsystem level, data were obtained pertaining to herd structure, milk production and composition, animal body weight and condition score, quality and intake of feedstuffs, and sanitary and reproductive management.

To determine soil physical and chemical properties and soil losses in grazing and cropping lands, soil sampling and runoff measurements were carried out during the rainy season on two selected farms. Following the Benchmark sampling design (Pennock

Table 1

Farm size, livestock, livestock density and composition, milk production and labor characteristics of six dairy farms of Marcos Castellanos, Michoacán, Mexico, and of the reference farm for farm yield gap analysis. FB denotes family-based and SS semi-specialized. I, M and E denote intensive, medium-intensive and extensive farm systems.

System	Total land ^a (ha)	Owned land	Rented land	Livestock ^b (LU)	Livestock density (LU ha ⁻¹)	Cows ^c (Nr)	Milk production (kg cow ⁻¹ d ⁻¹)	Hired labor (Labor-day y ⁻¹)	Family labor
FBE	57.9	0.0	57.9	18.9	0.33	12	7	0	548
FBM	46.7	11.2	35.5	35.2	0.75	24	10	185	365
FBI	23.6	23.6	0.0	31.2	1.31	18	14	387	148
SSE	95.1	95.1	0.0	59.2	0.62	42	10	1110	114
SSM	132.8	101.0	31.8	141.7	1.06	92	9	1208	365
SSI	23.6	12.3	11.3	37.3	1.57	27	14	1000	532
Reference	29.8	–	–	29.8	1.00	25	14	–	–

^a Owned and rented land used for grazing and cropping.

^b Livestock units (LU) are defined in Mexican terms as a 450 kg LW lactating cow with a calf younger than 7 months.

^c Total number of lactating cows in herd.

et al., 2006), two 30 × 30 m² plots in grazing land and two in cropping land were sampled monthly. One soil sample was constituted by 30 subsamples randomly taken in each plot with the soil sampler buried up to 25 cm deep. Beside every plot, three runoff plots of 1 × 5 m² were installed to capture runoff after each precipitation event (Hudson, 1993).

A direct method of forage maize sampling was implemented to estimate forage maize yield (Crespo-Lira et al., 2007) previously harvested for silage or “stover” (local name for ground dry whole maize plants including cobs). In every parcel of forage maize on each farm, the number of maize plants in 2 m row length was counted and cut at 18 cm above ground level in 20 random samples. To measure the DM content and the nutritional value three samples were taken from each of the different forage types, i.e. fresh forage (after weighing and chopping) and silage and stover (at the end of the conservation process).

Because availability and management of feed differed importantly between dry and rainy seasons, dry matter intake (DMI) was analyzed for each season separately. For the dry season, DMI was calculated from the amount of purchased feedstuffs and conserved forage provided, such as concentrates, maize silage, stover and occasionally sugar cane tips, as reported by the farmer. For verification, DMI was measured directly once a month. During the rainy season, total DMI and forage intake and production from grazing lands were estimated based on the ME requirement of lactating cows (Baker, 2004). DMI from concentrates and conserved forage were estimated in the same way as in the dry season. Since no feed losses in the feed troughs were found, feed utilization was considered to be 100%. Intake from grazing was estimated by deducting the ME taken in through concentrates and conserved forage from the total ME required for cattle, and computing forage production assuming 62% of forage utilization (Smart et al., 2010). This approach was used because of the difficulty of accurately measuring DMI of cattle when intake is composed of diverse grassland species and browsed shrubs and trees in a continuous stocking system. To estimate the nutrient composition of grazed forage, a monthly hand-plucked sample was analyzed. The parameters to estimate the requirements of DMI, ME and CP per age category of cattle were based on NRC (1989, 2001). For lactating cows the ME requirements the original equation was based on an adjusted for equation lactating cows in for grazing systems in Mexico (Améndola, 2008):

$$\text{ME} = (\text{MW} \times \text{BM}) \times \text{GCF} + \left(\text{LW} \times 0.1 \times \frac{\text{CWCh}}{\text{DWG}} \right) + (\text{MF} \times \text{FC} + 1) \times \text{MY} \quad (1)$$

where ME is metabolic energy (Mcal d⁻¹), MW metabolic weight (LW/0.75), BM basal metabolic factor (0.127), GCF grazing cost factor estimated as 1.31 for the dry season and 1.34 for the rainy season to

account for the differences in length of the two seasons, LW live weight (kg), CWCh cost weight change factor (1.45), DWG daily weight gain (g d⁻¹), MF milk fat (%), FC milk fat cost factor (0.165), I intercept of the equation for calculating ME requirement per liter (0.577) and MY milk yield (L d⁻¹).

Monthly, quantitative farm information was collected on: milk used on the farm and sold, milk production per cow, number of dry and lactating cows, number of births, number of cows in heat, and purchase of inputs. Individual food ration were measured, body condition of the cows was visual estimated and mastitis incidence was established by the California mastitis test. Ten feces and urine samples were collected for chemical analysis.

N, P, K and total ash content were established by chemical analysis of samples of grazed and conserved forage, concentrates, milk, feces, urine, runoff, and soil. In the soil samples taken at beginning and at the end of the rainy season also bulk density and organic carbon were measured. ME content of feedstuffs was estimated through acid detergent fiber content (ADF) using the equation $\text{ME} = 3.412 - 0.0322 \times \text{ADF}$ (Table 15.5 in NRC, 2001). The CP content in all feedstuffs was estimated using the macro-Kjeldahl method (Harris, 1970).

2.4. Farming system analysis

The FarmDESIGN model (Groot et al., 2012) was used to quantitatively evaluate the performance of the selected dairy farms. The model describes a farming system in terms of annual balances of N, P, K, organic matter, labor and money. Separate balances of animal DMI, ME and CP were calculated for the dry and the rainy seasons to account for the major differences in feeding strategies. During the dry season, from December to June, the animals grazed but the amount of herbage ingested was small and almost all the DMI came from concentrates and conserved forage (Crespo, 2009). During the rainy season animals were grazed. The model was parameterized using the information collected at the individual farms. Model outputs provide insight into the current situation of each dairy farm, and help to identify key problems and production constraints.

2.5. N, P and K balances

Nutrient balances were developed for the farm as a whole and for the soil component (Fumagalli et al., 2011). At the farm level, N, P and K balances were calculated as the difference between nutrient inputs and outputs at the farm gate. Inputs included purchased fertilizers and animal feed, biological and non-symbiotic fixation, and atmospheric deposition whereas sold maize forage, animals, and milk were considered as outputs. To estimate the N, P and K flows associated with animal production, total DMI of each

feedstuff, sold milk and animals, and maize forage were multiplied by their corresponding N, P and K contents.

The nutrient balances for the soil component were based on the differences between soil additions and removal of N, P and K. Additions included incorporation of forage residues, biological and non-symbiotic fixation, atmospheric deposition, fertilizer application; removal comprised crop nutrient uptake, volatilization and erosion (Whitehead and Raistrick, 1990; Patra et al., 1996; Kirchmann and Lundvall, 1998). The soil N, P and K balances were calculated for cropland and grazing land separately to obtain an area-weighted average for calculating values at the farm level. Based on results of Cleveland et al. (1999) and Galloway et al. (2008) values for N-fixation, non-symbiotic N-fixation, and N-deposition were taken to be 3.5, 3.0 and 7.0 kg N ha⁻¹ y⁻¹, respectively. Values for P and K deposition at regional level were assumed to be 1 kg P ha⁻¹ y⁻¹ and 2 kg K ha⁻¹ y⁻¹ (Smaling and Fresco, 1993).

Nutrient surpluses were converted into monetary value by calculating the farm-specific cost per unit N, P and K based on amounts and prices of the inorganic fertilizers used on each farm and their nutrient contents. The monetary value of the surpluses was found by multiplying the nutrient-specific value by the surplus of corresponding nutrient and summing values for N, P and K.

2.6. Farm yield gap analysis

Farms may differ in their management of crops, animals, manure and soils. These subsystems and their interactions determine production and environmental performance. To reduce costs and environmental impact and to increase self-reliance, farms could improve utilization of their own feed resources, feed the herd in a balanced manner to maximize feed use efficiency, and use the produced manure to support soil fertility. In addition, herd structure could be balanced in terms of the ratio between productive and non-productive individuals. Yield gaps, i.e. differences between current and attainable production levels, may occur for both crop and animal subsystems. Since these yield gaps at subsystem level affect each other, they have to be considered at the appropriate level of integration, i.e. the whole farm.

We expressed the farm yield gap as the difference between milk production on a real farm and the milk production attainable on a virtual 'reference' farm with the same area and with the same number of lactating cows as the real-world 'counterpart'. The virtual reference farm was constructed by combining best performing crop and animal production activities as observed among the 6 case study farms. We assumed that rangeland and improved pastures were used as grazing resources, and that maize was fed as silage instead of stover (Table 2). Production data for maize and improved pasture came from farm FBI, for rangeland data from farm SSM were used. The maize crop of farm FBI was fertilized at a rate of 180 kg N/ha. While the number of cows on each reference

farm was based on those observed on the real 'counterpart', milk yield per cow was derived from measurements on the best performing farm (SSI) (Table 1): 14 kg cow⁻¹ d⁻¹ with a protein and fat content of 3.3% and 3.4%, respectively. We assumed improved herd management with an annual cow replacement rate of 25%, and equal distribution of young animals between calves and yearlings. Milk production was assumed to start at the age of 2 years, in agreement with the 'best practices' assumption for the reference farm. Feed rations were calculated by maximizing milk production per hectare of on-farm feed, conditional on maximum feed intake and less than 5% deviation from equilibrium feeding for ME and CP, using the optimization routines in the FarmDESIGN model (Groot et al., 2012). From the set of possible combinations of rangeland, improved pasture and maize silage that resulted in feasible rations, we selected one in which maize occupied 25% of the total feed area, similar to the largest proportion found in the set of case study farms (23.3% on farm SSI). The remainder of the area of the reference farm was occupied by improved pasture (31%) and rangeland (44%) (Table 2). The on-farm produced feeds were able to support a milk production level of 2.7 Mg ha⁻¹, which was limited by protein supply rather than metabolizable energy availability. Lifting the protein shortage by assuming import of CP with supplementary feed resulted in a productivity of 4.3 Mg ha⁻¹. This attainable production level was used for whole-farm comparison of nutrient balances between the reference farms and the real farms.

Partial farm yield gaps were calculated to attribute the milk yield gap to the 3 major differences between a reference farm and its real-world counterpart: differences in herd management and livestock density, differences in proportions of cultivated crop areas and differences in crop productivity. The contribution of each component was calculated by sequentially changing the values used for the virtual farm to those of the real farm.

3. Results

3.1. Farm yield gaps for on-farm feed production

The milk production intensity increased strongly with increasing livestock density (Fig. 1a). However, when considering the milk production that could be attributed to on-farm produced feed resources, milk production per unit of area increased more slowly, indicating that a large part of the increase in milk yield was related to enhanced inputs of external feed resources and the self-reliance for feeds declined with higher livestock density.

The farm yield gap analysis demonstrated that at a livestock density of 1.0 LU ha⁻¹ a milk production level of 2.7 Mg ha⁻¹ would be attainable based on on-farm produced feeds, whereas with supplementary feed imports the production level of 4.2 Mg ha⁻¹ could be attained. The farm yield gap for milk production solely based on

Table 2
Reproduction and lactation characteristics, herd structure and herd health status on six farms representing extensive (E), medium-intensive (M) and intensive (I) family-based (FB) and semi-specialized (SS) dairy systems in Marcos Castellanos, Michoacán, Mexico, and of the reference farm for farm yield gap analysis.

System	Calving interval (days)	Lactation duration (days)	Herd structure (%) ^a			AI ^b (%)	Mastitis ^c (%)
			Dry cows	Calves	Heifers		
FBE	438	353	19	33	67	0	34
FBM	474	352	26	29	58	37	54
FBI	356	320	10	39	83	27	77
SSE	499	337	32	26	52	99	60
SSM	438	277	37	30	61	40	75
SSI	438	365	17	22	48	95	59
Reference	365	305	16	35	35	–	–

^a Number of dry cows, calves and heifers expressed as a percentage of the total number of milk cows (lactating and dry).

^b Cows artificially inseminated as a percentage of all cows served throughout the year.

^c Cows with mastitis as a percentage of cows sampled in the dry and rainy seasons.

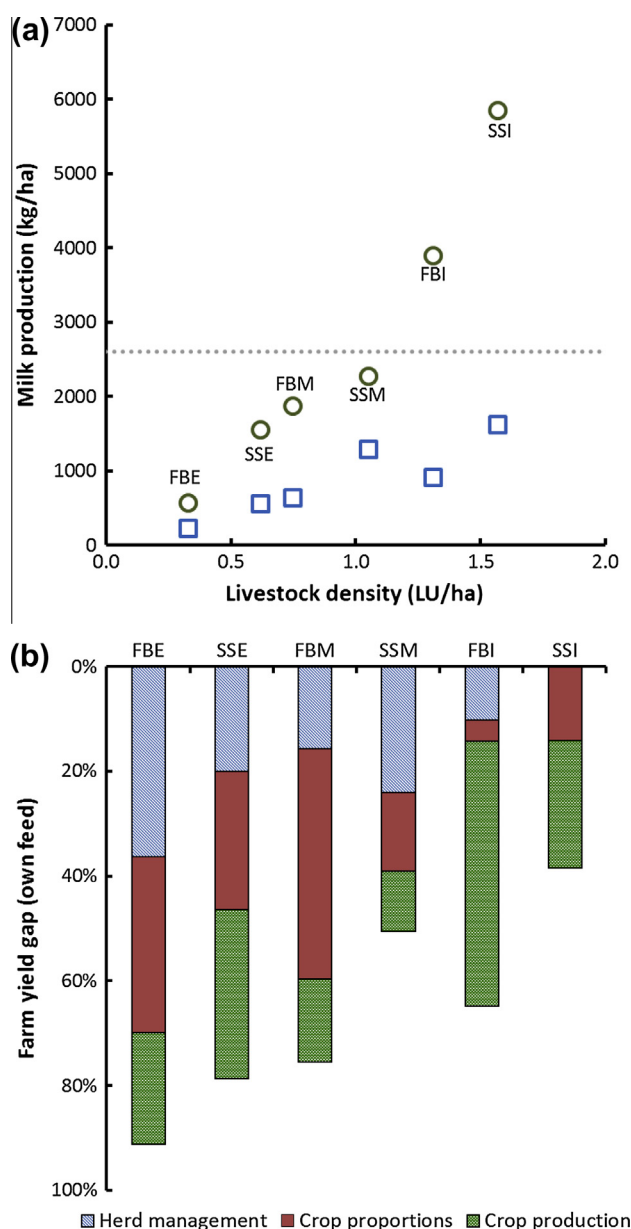


Fig. 1. (a) Milk production intensity in relation to the livestock density on different farms from own feed resources (□) and from total rations including imported feeds (○). The dashed line indicates the reference farm production level from own feeds at an optimal livestock density of 1.0 LU/ha. (b) Farm yield gap of case study farms compared to reference farms, and proportions of the gap that can be attributed to deficiencies in herd management, crop proportions and crop productivity.

on-farm feed resources between the reference farms and the case study farms ranged from 41% to 92% (Fig. 1b). This yield gap declined with higher livestock density and was on average higher for family-based farms (78%) than for semi-specialized farms (58%). Although the overall average reduction could be attributed almost equally to shortcomings in herd management (18%), allocation of crop proportions (23%) and lower crop productivity (26%), the causes of the yield gap differed strongly among farms (Fig. 1b). These will be analyzed in more detail below.

3.2. Herd management

The dominant breed on all farms was Holstein Frisian, but crosses of this breed with Zebu, Brown Swiss, and Jersey were also present. Average milk production was 4310 ± 103 kg cow⁻¹ lactation⁻¹ and

the milk production rate was slightly higher and less variable in the dry season (from October to May) than in the rainy season (Fig. 2).

Clear indications of herd management problems were observed (Table 2). The herd structure was imbalanced as the percentage of dry cows was higher than 16% (assuming a lactation period of 305 days and a minimum calving interval of 365 days) and the number of heifers kept for replacement was high. The large numbers of dry cows and heifers (Table 2) is an indication of inadequate fertility management caused by for instance lack of identification of cows in heat, poor implementation of artificial insemination, low quality of semen or deficient postpartum animal health care, whereas the body condition score was within the acceptable range (between 2.8 and 3.5; Table 4). The prevalence of mastitis varied from 34% to 77% of lactating cows at any one time. The high proportion of unproductive animals and the occurrence of mastitis reduced the overall productivity of the herd and resulted in high feed costs for maintenance of non-productive animals. For farm SSI there was no yield gap relative to the reference production level that could be attributed to herd management, because the herd characteristics of this farm were adopted for the reference farm. The high number of heifers of farm SSI was compensated by a smaller number of calves than the reference farm and the proportion of dry cows (and thus lactation length and calving interval) was the same as for the reference farm. The yield gap related to herd management (Fig. 1b) was largest for farms FBE (38%), SSE (22%) and SSM (26%). This was related to low livestock density (for FBE and SSE) and low milk production per animal (for FBE and SSM), whereas for all these farms the percentages of dry cows and replacement heifers were high (Table 1).

3.3. Crop management and animal feeding

On all farms predominant land-use was grassland, either semi-natural rangeland or improved pastures (Table 3). Only the most intensive farms (FBI and SSI) utilized improved pastures with Rhodes grass (*Chloris gayana*) and annual ryegrass (*Lolium multiflorum*), respectively. Forage maize covered between 5% and 23% of the farm area and was used for the production of stover and silage. In terms of forage production, the productivity of rangeland tended to be rather low (0.6 – 1.6 Mg DM ha⁻¹ year⁻¹) and although values for improved pastures were higher (2.8 – 3.4 Mg DM ha⁻¹ year⁻¹), the quality in terms of CP content was lower than for rangelands (Table 3). Maize yields ranged from 7.4 to 16.4 Mg DM ha⁻¹ among farms (Table 3). Low maize yields were associated with low plant density due to low seeding rates and/or poor initial crop establishment. After maize harvest stover production resulted in larger losses of CP and ME than ensiling.

The yield gap related to crop management (proportions and yields) ranged from 27% to 60% when compared to the reference production level (Fig. 1b). For the most intensive farms (FBI and SSI) this gap was caused mainly by low crop productivity, for farm FBI the productivity of rangeland was low and a large fraction of the maize was exported, and for farm SSI maize yields were low and maize was conserved as stover, resulting in stronger quality reduction of the harvested material than for silage (Table 3). Farms FBI and SSI used 20–24% of the surface area for maize cultivation and a large proportion of their surface area as improved grasslands, so the impact of crop proportions was only marginal. For the other farms crop proportions contributed to the farm yield gap (15–44% yield reduction relative to the reference production level; Fig. 1b) due to low proportions of maize cultivation and smaller areas of improved pasture (Table 3). Moreover, the crop production levels of rangeland and maize on these farms were low, except for rangeland of farm SSM that was used to construct the reference farm, so

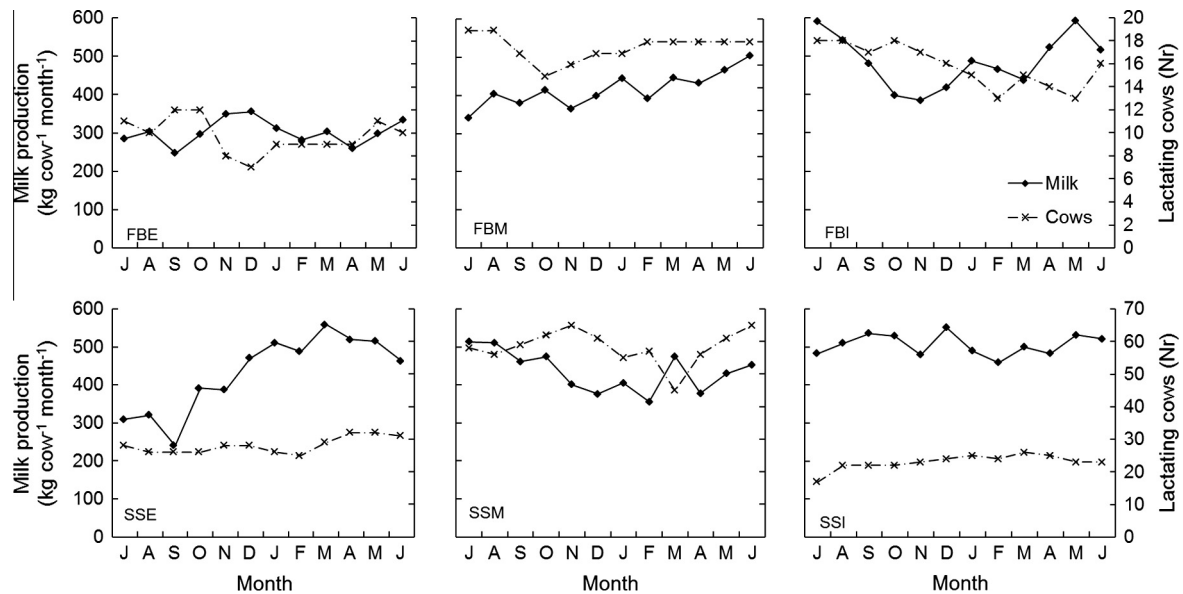


Fig. 2. Monthly milk production per lactating cow (broken line) and number of lactating cows (solid line) in 2009/2010 on six farms representing extensive (E), medium-intensive (M) and intensive (I) family-based (FB) and semi-specialized (SS) dairy systems in Marcos Castellanos, Michoacán, Mexico.

Table 3

Crop and grassland area, productivity and quality for six farms representing extensive (E), medium-intensive (M) and intensive (I) family-based (FB) and semi-specialized (SS) dairy systems in Marcos Castellanos, Michoacán, Mexico, and the reference farm for farm yield gap analysis. Shaded areas indicate crops selected for reference farm construction.

Variable	FBE	FBM	FBI	SSE	SSM	SSI	Reference
<i>Grazing land^a</i>							
Type ^b	R	R	R	IP	R	R	IP
Area (ha)	54.6	44.5	12.8	5.3	76.0	7.5	109.6
Yield (Mg DM ha ⁻¹)	0.78	1.01	0.63	2.77	1.32	1.06	1.60
ME (Mcal kg ⁻¹)	2.1	2.3	2.3	2.2	2.3	2.3	2.3
CP (g kg ⁻¹)	114	117	117	90	114	107	147
<i>Forage maize</i>							
Area (ha) ^c	2.6	2.2	5.5	11.6	23.2	4.5	7.5
Yield (Mg DM ha ⁻¹)	10.3	16.0	16.4	7.7	16.3	7.4	16.4
Plant density (m ⁻²)	4.3	5.3	7.1	4.3	5.6	5.0	7.1
Maize on-field cost (US\$ ha ⁻¹) ^d	331	527	680	868	646	566	680
Maize on-field plus ensiling cost (US\$ ha ⁻¹)		807	1105	1139	968		1105
Maize on-field plus stover production cost (US\$ ha ⁻¹)	478				1200	916	
ME _{harvest} (Mcal kg ⁻¹) ^e	2.2	2.0	2.1	2.3	2.4	2.2	2.1
ME _{silage} (Mcal kg ⁻¹) ^e		2.7	2.5	2.6	2.3		2.5
ME _{stover} (Mcal kg ⁻¹) ^e	1.6				2.1	2.6	
CP _{harvest} (g kg ⁻¹) ^e	79	116	107	78	88	95	107
CP _{silage} (g kg ⁻¹) ^e		80	96	78	81		96
CP _{stover} (g kg ⁻¹) ^e	56				76	66	

^a DM: dry matter, ME: metabolic energy, CP: crude protein.

^b Grazing land types: IP: improved pasture, R: rangeland.

^c Net area cultivated.

^d Cost of forage maize production until harvest.

^e Metabolizable energy (ME) and crude protein (CP) of maize at harvest, or after ensiling and stover production, respectively.

that farm SSM exhibited a crop production related yield gap of only 11% due to lower maize yield than the reference farm.

3.4. Rations and seasonal variation

Roughage constituted the main component of the ration during both seasons, the proportion in the rainy season slightly exceeding that in the dry season (69% vs. 55%) (Table 4). During the rainy season grazing was the main source of forage whereas in the dry season conserved forage produced on the farm and concentrates purchased locally were provided. The proportion of concentrates in the ration at herd level varied between 20% and 34% in the rainy season and between 30% and 73% in the dry season (Table 4). Dependence on

purchased foodstuffs was similar in both seasons for the two farms with a livestock density of ca. 1 LU ha⁻¹ (i.e., FBI and SSM), whereas farms with lower (FBE, FBM and SSE) and higher (SSI) livestock density demonstrated less consistent ration composition and strongly enhanced feed imports in the dry season (Fig. 3).

Differences between seasons in feed sources, composition of the rations and external dependence influenced feeding costs (Table 4). Feeding costs for the whole herd were 63% higher during the dry season than during the rainy season and 41% for lactating cows. Deficiencies in crop and feed management and in herd management that were revealed in the farm yield gap analysis were reflected in ration use efficiency expressed in kg milk per unit of DMI (Fig. 4a) and in feed costs per kg milk produced (Fig. 4b).

Table 4

Animal intake, ration composition, ration nominal costs and animal body condition score per season of the herd on six dairy farms representing extensive (E), medium-intensive (M) and intensive (I) family-based (FB) and semi-specialized (SS) dairy systems in Marcos Castellanos, Michoacán, Mexico. Data were collected in 2009/2010.

Variable	Rainy season ^a						Dry season ^b					
	FBE	FBM	FBI	SSE	SSM	SSI	FBE	FBM	FBI	SSE	SSM	SSI
Intake (kg DM LU ⁻¹ d ⁻¹) ^b	10.6	8.4	9.7	13.0	11.1	13.6	8.4	8.8	9.6	10.1	10.5	14.5
Intake (kg DM 100 kg ⁻¹ LW) ^c	2.34	1.84	2.36	2.73	2.40	2.92	1.37	1.97	1.95	2.36	2.44	3.39
Forage (g kg ⁻¹ DMI)	802	699	512	665	729	733	269	589	435	650	704	630
Concentrates (g kg ⁻¹ DMI) ^d	198	301	488	335	271	267	731	411	565	350	296	370
Feeding cost (US\$ kg ⁻¹ DM)	0.08	0.14	0.22	0.11	0.13	0.15	0.29	0.23	0.27	0.27	0.14	0.15
Body condition score	3.0	3.0	3.4	2.8	2.9	3.5	2.8	2.9	3.3	3.0	3.1	3.5

^a The “rainy” season pertains to the period from July to September when pastures provide a significant contribution to the ration; during the “dry” season, from December to June, there is very limited grazing.

^b Livestock unit (LU) is defined in Mexican terms as a 450 kg LW lactating cow with a calf younger than 7 months.

^c The DMI expresses the intake as a percentage of the body weight (kg LW).

^d Concentrate values represent their content (g) per kilogram of DMI.

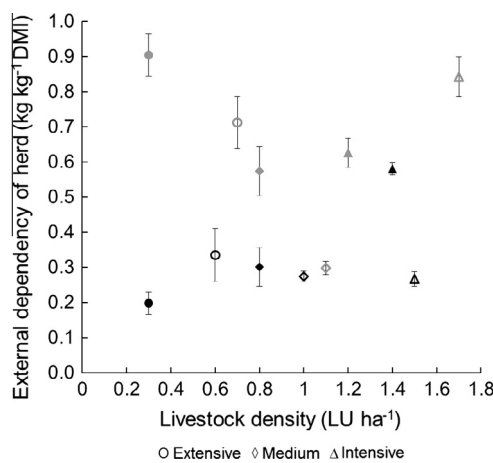


Fig. 3. Relationship between livestock density and external feed dependency (calculated as the proportion of imported feeds in the animal ration) in 2009/2010 on six dairy farms representing extensive, medium-intensive, and intensive family-based (FB, open symbols) and semi-specialized (SS, solid symbols) dairy systems in Marcos Castellanos, Michoacán, Mexico. Black and gray symbols denote rainy and dry season, respectively.

3.5. Nutrient balances

The input of nutrients at the farm gate (kg ha⁻¹) increased with milk production intensity (kg milk ha⁻¹). Major inputs of N and K came from bought feedstuff (Fig. 5a and e), while inorganic fertilizer was the main source of P (Fig. 5e). The N from feed inputs

increased linearly with production intensity, whereas fertilizer inputs tended to level off (Fig. 5a) because fertilizer inputs were relatively low for SSI compared to FBI. Milk was the main N output, but in some instances farms also exported substantial amounts of nutrients via crop products (Fig. 5b). This was of particularly evident for FBI that exported 31 kg N ha⁻¹ as silage maize. On average, N, P and K surpluses were 75 ± 16, 15 ± 6, and 19 ± 6 kg ha⁻¹, respectively. The total N surplus increased with production intensity (Fig. 5c), although the N surplus per unit of milk decreased (Fig. 5d). Surpluses of P and K tended to increase with production intensity, although the relation was less pronounced than for N (Fig. 5e). The total N surplus was nearly identical to the amount of N excreted via urine and manure (Fig. 5f), which showed that shortcomings in manure handling are a primary cause of nutrient losses.

Nutrient balances of the reference farms and their real-world counterparts were compared. The N inputs with supplementary feeds needed to reach the target yield of 4.3 Mg milk ha⁻¹ for the reference farm were substantially lower than the trend for the real farms (Fig. 4a), but the N surpluses per ha and per unit of product were only slightly lower than the trend for the real farms (Fig. 5c and d). This is attributable to the fertilizer input of 180 kg N ha⁻¹ for cultivation of the maize crop that was adopted on the reference farms.

4. Discussion

In this study we estimated the yield gap at farm level, by comparing actual farm performance to that of virtual reference farms for which we assumed best practices observed in the case study

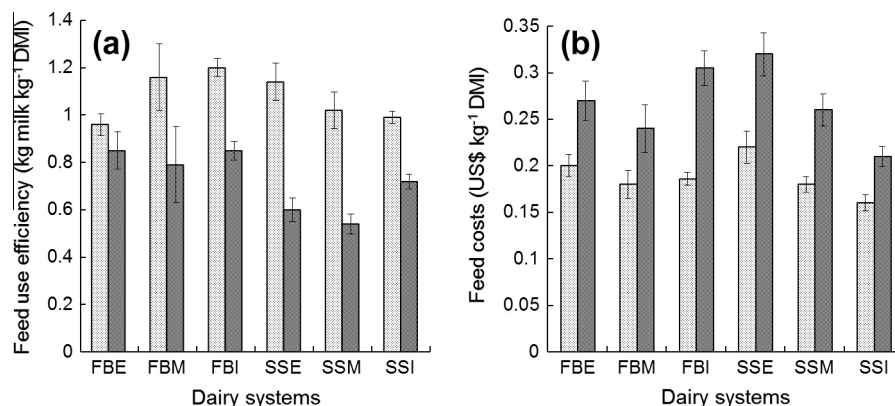


Fig. 4. Feed use efficiency (a.) and feed costs (b.) for lactating cows (light gray) and at herd level (dark gray) on six dairy farms representing extensive (E), medium-intensive (M) and intensive (I) family-based (FB) and semi-specialized (SS) dairy systems in Marcos Castellanos, Michoacán, Mexico.

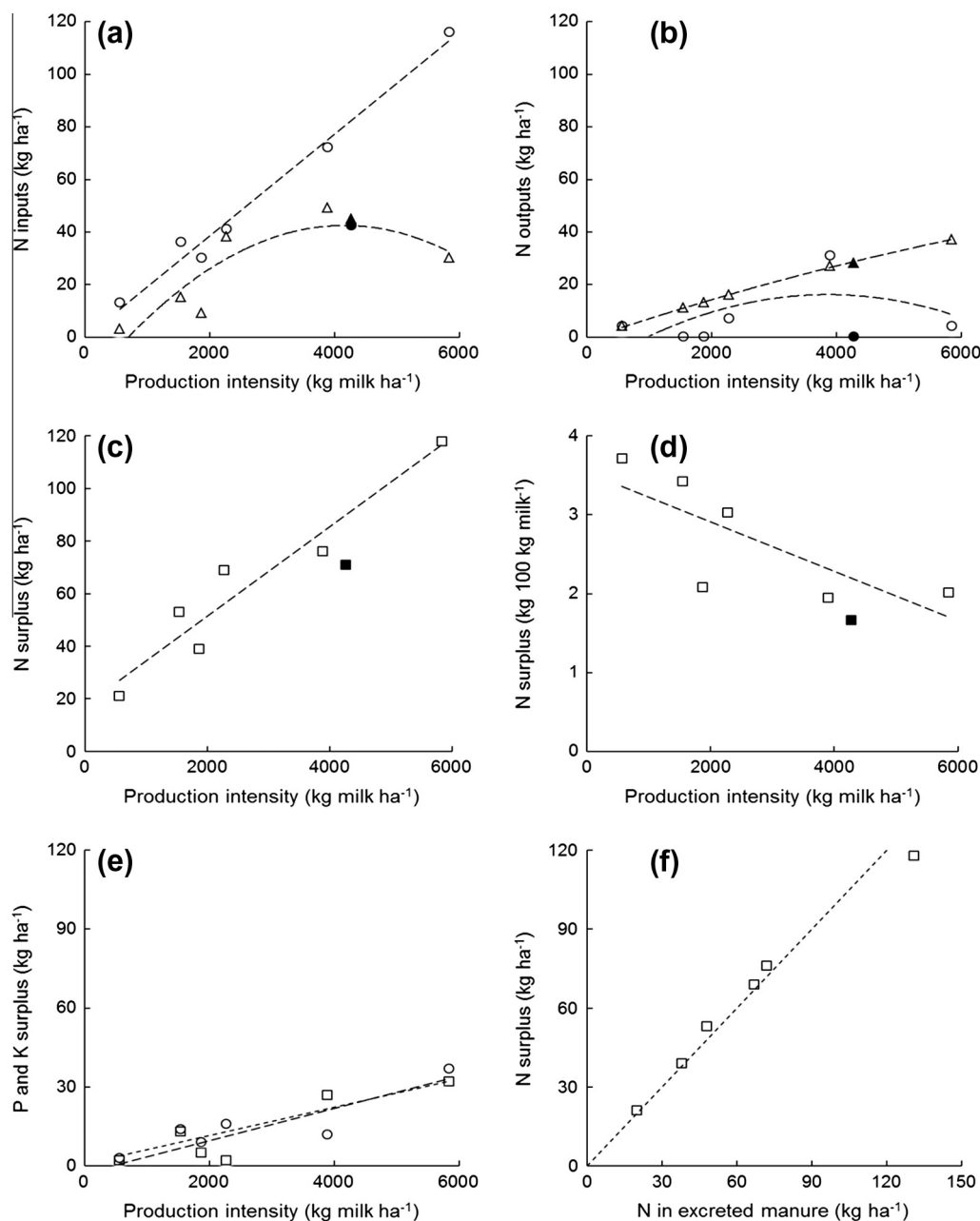


Fig. 5. Nutrient use indicators of 6 case study dairy farms in North West Michoacán as a function of production intensity (a–e) and manure production (f). (a) N inputs (fertilizer Δ , feed $\circ$); (b) N outputs (milk and meat Δ , and crop products $\circ$); (c) N surplus per unit of farm area (P $\square$, K $\circ$) as a function of production intensity (kg milk ha⁻¹); (d) N surplus per unit of product; (e) P and K surpluses per unit of farm area; (f) N surplus as a function of excreted manure. Solid symbols denote values used for the reference farm. Dashed lines indicate average trends. Inputs from symbiotic and non-symbiotic N fixation and deposition are not shown, but were included in the surplus calculations.

region to construct attainable yield levels based on-farm and imported feed resources. The farm yield gap analysis pointed to management constraints in the crop, animal and manure components of the farm systems. Although the values of the model parameters were collected directly on the case study farms, the analysis was based on a number of simplifying assumptions, which call for prudent interpretation of the results. The reference farms were based on maximizing feed self-reliance in terms of ME and CP, but did not consider objectives concerning family income, operating profit, labor, nor the uncertainty in land tenure. A single yield level was assumed for maize, rangeland and improved pastures; variation between fields and farms was not taken into account. The options for on-farm feeding were based on what was practiced in the region. It will be highly desirable to investigate a

broader range of options, including those with innovative feed sources such as grass-legume pastures or protein banks that can help to close the CP gap. And finally, farmer skills and abilities were not considered in the construction of the reference farm. Our observations, however, suggested that changes in the production systems should be accompanied by improvement of managerial abilities (cf. Groot et al., 2006). For some farmers projected increases in crop and milk yields were considerable (Fig. 1; Table 2). Such improvements could only be attained by management changes throughout the farm based on a clear strategy, focusing on better reproductive, sanitary, and feed management, since the calving interval, age at first calving, prevalence of mastitis, and external dependence on feedstuffs have to be reduced to close the gap (cf. Table 2).

The three dimensions of farm management that determine the yield gap (herd management, crop proportions and crop productivity) were assumed to be additive. In reality there will be interactions, but because we focus primarily on effects of feeding strategy and on milk yield as dominant response variable this seems justified, since milk production responds quickly to feeding. An exceptional phase in this respect is early lactation, wherein feed deficiencies could be compensated by mobilization of body reserves resulting in decline in body condition score. However, the farmers fed cows in early lactation sufficiently so that body condition scores did not decline considerably. Moreover, we were particularly interested in the relative importance of the three determining management aspects, which is well reflected even when assuming additive effects.

Herd management and forage availability in the course of the year dictated to a large extent the overall productivity and profitability of the studied dairy systems. The observed shortcomings in health care and reproductive management of the herd were the most likely cause for the unbalanced herd structure and low milk production per cow. Insufficient forage production in grazing lands and cropland caused a strong dependency on external feed (Fig. 3), which affected overall profitability negatively and translated into important nutrient surpluses, which potentially cause negative environmental impacts at both the local and regional level. The farm yield gap analysis showed that better crop and herd management could improve the self-reliance of farms considerably, although additional adjustments on the farms would be needed to improve whole-farm nutrient use efficiency and reduce nutrient losses.

Milk yield revealed inefficiencies associated with herd structure and management (Table 1; Fig. 4). When considering only lactating cows, milk production by both the family-based and the semi-specialized dairy systems was similar to the regional averages of respectively 3800 and 4500 kg cow⁻¹ lactation⁻¹ reported in the literature (Amendola, 2002; Espinoza-Ortega et al., 2005). The level and quality of animal nutrition is one of the most important production constraints in smallholder livestock systems (Thornton, 2010). In North-west Michoacán there are two contrasting feed management strategies associated with distinct seasonal rainfall patterns, namely open grazing during the rainy season and supplementary feeding during the dry season. During the rainy season, animals grazed mostly on rangelands, which consist of 107 different species of the native flora distributed at regional scale (Cortez-Arriola, submitted for publication). Although SAGDER (2000) reported that in this region rangelands can typically hold stocking rates of 0.13 LU ha⁻¹, we observed much higher livestock density in the case study farms (Table 1). Replacing rangelands with improved pastures can increase forage production by about 200% (Table 3), thereby reducing farm dependence on external feedstuff during the rainy season.

During the dry season, rangeland-based roughages were replaced by conserved maize forage (silage maize and stover), alfalfa hay, and ground sugar cane tips. None of the farms produced enough conserved forage to replace grazing, resulting in a larger share of bought products (Table 4; Fig. 3). Nevertheless, feed balance calculations showed that shortage of CP in the dry season ration was likely on 4 out of 6 farms as well as on the reference farms. Potential on-farm alternatives to overcome this shortcoming include the establishment of improved grass-legume pastures or protein banks (Ibrahim and Mannetje, 1998; Arriaga et al., 2000). Calculations indicated shortage of ME on one farm at herd scale, which may be a further indication of the need to improve management skills.

The studied dairy systems were characterized by appreciable N, P, and K surpluses, which points to inefficiencies, high potential losses and/or environmental emissions (Koelsch, 2005; Powell

et al., 2010). Sotelo et al. (2005), Ramírez et al. (2007) and Silva-García et al. (2006) reported that agricultural activities developed in the “Chapala” sub-basin cause negative environmental impacts for Lake Chapala and the surrounding areas. Large loading rates of N, orthophosphates, and total P compounds originating from fertilizers and wastewater with manure from agricultural areas are mentioned as causes of eutrophication. This study shows that there is scope for improvement of both environmental and economic performance of farms by implementing tactical and strategic system changes (cf. Shah et al., 2013).

The high surpluses appear to result first of all from poor manure management. Most dairy farms in the study region have implemented no measures for collecting and re-cycling dairy manure within their production system. Manure is perceived as a waste product that costs money and labor to be removed, rather than a valuable resource that can help to sustain soil fertility. One of the key issues is that due to poor grazing management, manure tends to accumulate in small areas where it leads to high gaseous emissions, soil degradation, and nutrient loading of surface waters due to runoff (Van der Molen et al., 1998). Changes in grazing management could promote more even manure deposition, reduce emissions, and support more effective utilization by closing nutrient cycles. Cycling dairy cattle over different areas and land-use types can increase manure-based NUE by 30–50% (Powell et al., 2006; Powell et al., 2010).

5. Conclusions

Through integrated assessment of representative farms with different intensification levels we revealed key differences in farm management practices that impact on the productivity and profitability of dairy systems in NW Michoacán. The results of our analysis, pointing to large dependence on external inputs, poor internal efficiencies and acute environmental externalities, suggest that current dairy farming in the region is unsustainable. Small dairy farms are caught in an intensification trap that results from the implementation of standard technological packages with poor adaptation to local conditions – notably to the long dry season. A number of strategies and management practices developed by some local producers were identified that may help other farmers avoid the intensification trap. These were evaluated in a farm yield gap analysis to account for interactions between field and animal level changes. Improved production and nutrient use efficiencies were shown to result from more attention for on-farm feed production for the rainy season and particularly for the dry season, more attention for balanced feed rations, and more attention for cattle management to reduce inter-calving periods and improve cow health. In addition to these changes, avoiding local excess of plant nutrients by better manure management may be expected to benefit both environmental and economic performances. Common denominator in the changes is reducing external nutrient inputs by increased farm-internal nutrient cycling.

Acknowledgements

This research was funded by the EULACIAS project (EU FP6-5832004-INCO-dev-3; contract nr 032387). We thank the dairy farmers of Marcos Castellanos, specially to Arturo Padilla, Daniel Chávez, Everardo Haro, Filiberto García, Honorato González, and José Luis Orozco for their cooperation and willingness to be part of this project. We are grateful to our colleagues Ramés Salcedo Vaca, Máximo Huerta Bravo, Luis Humberto Valencia Chávez, Efrén Trujillo López, Isela J. López Herrera, and Heriberto Estrella Quintero for their important support during field work and in project management.

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